

Institute Name:

Malla Reddy College of Engineering for women

Title of the Innovation/Prototype:

Health Monitoring on Social Media Over Time

Team Lead Name:

Jahnavi Avadhuta

Team Lead Email:

jahnavijanu666a@gmail.com

Team Lead Phone:

9392789103

Team Lead Gender:

Female

FY of Development:

2024-25

Developed as part of:

Academic Requirement/Study Project

Innovation Type:

Service

TRL LEVEL:

6

MRL Level:

MRL 3: Manufacturing proof of concept developed

IRL Level:

IRL 3: Validate Problem - Solution Fit (Confirmed Value Proposition & Techno-Economic Analysis) & Minimum Product Cost (Maturity of Core Technology)

Theme:

Healthcare & Biomedical devices.,

Define the problem and its relevance to today's market / society / industry need:

Despite a surge in health-related content on social media, there is no standardized system to effectively analyze and monitor this data over time. Health organizations lack tools to convert noisy, unstructured content into actionable insights for timely decision-making. Key challenges include identifying relevant posts, handling slang and sentiment shifts, tracking trends across time and regions, and ensuring user privacy. This project aims to develop a scalable and ethical system that leverages machine learning and NLP to analyze health discourse on social media, extract meaningful indicators, and visualize evolving trends for improved public health monitoring.

Describe the Solution / Proposed / Developed:

The proposed solution is a scalable system that uses machine learning and natural language processing (NLP) to analyze health-related social media posts in real time. It collects data from platforms like Twitter and Reddit, filters and preprocesses it to remove noise, and extracts meaningful health indicators such as symptoms, diseases, and sentiments. The system tracks emerging trends over time and presents insights through interactive visualizations. It also ensures ethical data use by anonymizing user information, supporting privacy and responsible monitoring. This approach enables timely public health awareness, early intervention, and improved decision-making based on real-world health discourse.

Explain the uniqueness and distinctive features of the (product / process / service) solution:

The solution stands out by combining real-time social media monitoring with advanced NLP and sentiment analysis to extract meaningful health insights. Unlike traditional systems, it handles informal language, slang, emojis, and multilingual content, making it adaptable to diverse online expressions. It ensures user privacy through data anonymization and adheres to ethical standards. The system's ability to detect emerging health trends and visualize them through interactive dashboards enables timely interventions. Its scalability across platforms and regions makes it a powerful tool for modern public health surveillance, bridging the gap between unstructured online data and actionable health intelligence.

How your proposed / developed (product / process / service) solution is different from similiar kind of product by the competitors if any:

Unlike existing tools that rely on basic keyword tracking or delayed health reports, our solution uses advanced machine learning and NLP techniques to understand context, sentiment, and health-specific language in real time. It uniquely handles informal expressions, sarcasm, and multilingual content—capabilities most competitors lack. Additionally, it emphasizes ethical data use by anonymizing user information, which is often overlooked by others. The system provides dynamic, interactive visualizations of emerging health trends, making it more insightful and actionable than static reports. Its real-time analysis and scalability across platforms set it apart from conventional health monitoring solutions.

Is there any IP or Patentable Component associated with the Solution?:

No

Has the Solution Received any Innovation Grant/Seefund Support?:

No

Are there any Recognitions (National/International) Obtained by the Solution?:

No

***Is the Solution Commercialized either through Technology Transfer or Enterprise Development/Startup?:**

No

Had the Solution Received any Pre-Incubation/Incubation Support?:

No

Video URL:

<https://drive.google.com/file/d/1-4khJrKz3sx6FIT8whyNMkCrSs-nK5vE/view?usp=drivesdk>

Innovation Photograph:

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